

AgriAssist: A Decoupled, Agentic Multimodal Crop Advisory System

Milestone 2 Defense — Dataset Identification, Understanding & Production-Ready Preprocessing

SCOPE

Smallholder Farmers in Uttar Pradesh (Rice & Wheat Crops)

DELIVERABLE

4 Verified Data Streams Ready for Milestone 3 Model Training

Bridging the Lab-to-Field Gap: Architecture-to-Data Lineage

Mapping Our Four Data Streams Directly to Decoupled System Subsystems

Vision Disease Router (CNN)

Powered by a **Unified 20-Class Vision Dataset** containing 12,859 images.

12,859

Images (20 Classes)

Grounded Advisory Engine (RAG/LLM)

Fueled by **KCC Farmer Q&A** with 1.46M chunks and 170 authoritative policy PDFs.

1.46M

Chunks & 170 PDFs

Tool Execution & Forecasting

Driven by a **Harmonized Multi-Crop Yield Dataset** comprising 440,962 rows.

440,962

Multi-Crop Yield Rows

Data Governance, Open Licensing & Proactive PII Protection

Ensuring Full Ethical Compliance Across Public & Citizen-Facing Agricultural Data

Open Licensing

Utilizing 100% open research and public portals, including Kaggle CC-BY, data.gov.in, PPQS advisories, and Directorate of Economics & Statistics.

Automated PII Redaction

Executed a custom pipeline across 1.7M Kisan Call Center records, stripping phone numbers, emails, and farmer IDs.

Advisory Safety Floor

Explicit confidence thresholding and deterministic abstention configured to prevent hallucinated chemical recommendations.

Unified Vision Dataset: Harmonizing 12,859 Images Across 20 Classes

Repairing Noisy Labels to Build a Clean Rice & Wheat Foliar Disease Taxonomy

12,859

Unified Standardized Images

20 Biological Classes

Multi-Crop Pathogen Cohort

Consolidates highly diverse sources into a single standardized taxonomy for robust model evaluation.

- **15 Wheat Classes:** Covers critical rusts, blots, and pest damage.
- **5 Rice Classes:** Standardizes major bacterial and fungal pathogens.

Label Repair & Standardization

Resolving Raw Kaggle Data Corruption

Solved raw dataset vulnerabilities where Wheat folders were split into 45 noisy, redundant labels. Collapsed and mapped them into canonical biological classes to prevent extreme training dispersion.

Rare-Class Safeguard

Mitigating Severe Class Imbalance

Enforced a strict minimum evaluation floor for the highly limited 40-image rice leaf smut class. Integrated with weighted sampling mechanisms to ensure unbiased optimization during validation runs.

Eliminating Domain Shortcuts & Proving Zero Cross-Split Leakage

Why Naive Splitting Fails & How Group-Aware Perceptual Hashing Protects Generalization

⚠️ LEAKAGE DISCOVERY (CROSS-SPLIT COLLISIONS)

Severe Data Leakage Found in Raw Kaggle Splits:

- **Wheat Dataset:** Found ~44% near-duplicates
- **Rice Set 1:** Found ~80% near-duplicates
- **Identity Collisions:** 646 cross-split overlapping images

⚠️ DOMAIN SHORTCUT DISCOVERY

Non-Generalizable Features Detected:

- **Tungro Class Bias:** Identified 4:3 brown soil backgrounds unlike clean leaf close-ups in other classes.
- **Resolution:** Mitigated via letterbox padding and aggressive background augmentation.

🛡️ Technical Resolution & Validation

Group-Aware Stratified Splitting:

Implemented grouping based on **Perceptual Hash IDs** to isolate near-duplicates into identical splits, guaranteeing zero cross-split data leakage.

✓ **Verified 100% Mathematically Safe Split Boundary**

MATHEMATICAL SPLIT (ZERO LEAKAGE VERIFIED)



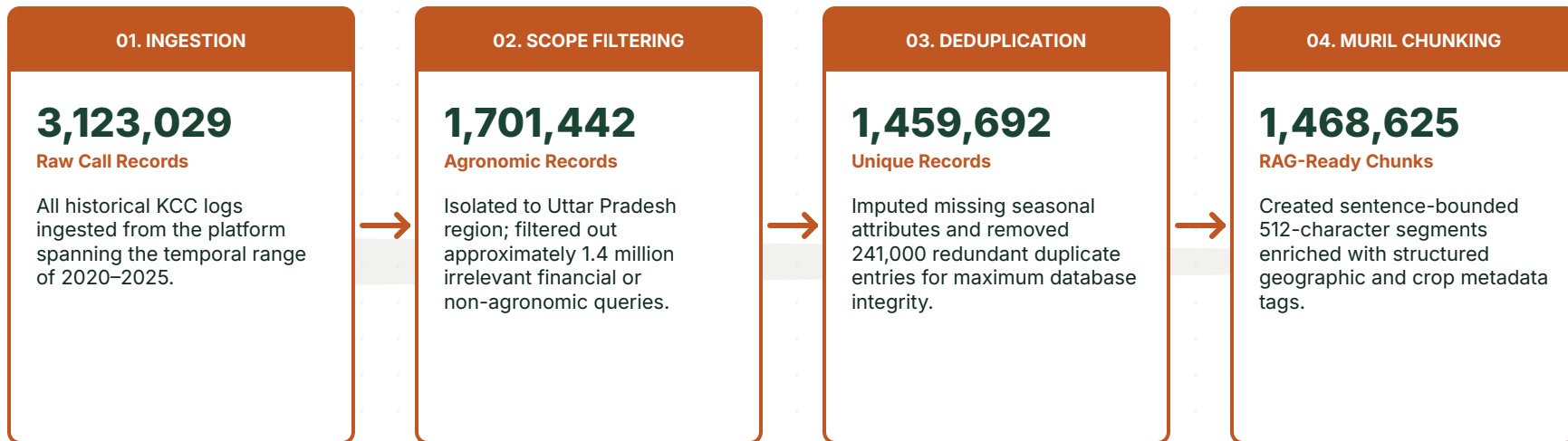
Train: 80%

Val: 10% Test: 10%

Strict stratification on hash group IDs protects generalization metrics

Kisan Call Center (KCC) Filtering: Curating 1.46M Agronomic Chunks

Distilling 3.12M Raw Logs into High-Signal Uttar Pradesh Advisory Knowledge



Authoritative Policy Grounding: 170 Government Agriculture PDFs

Complementing Experiential Call Logs with Quantitative Regulatory Standards

187

SOURCED

Exceeding Targets

Sourced 187 official PDFs including PPQS Advisories, UP Agriculture Contingency Plans, and ICAR Handbooks.

17

REJECTED

Quality Audit

Quarantined 17 corrupted or empty scanned files to ensure clean and parsing-ready source documents.

170

PRISTINE

Grounding Corpus

Resulting in 170 pristine, high-authority government agriculture regulatory PDFs fully prepared for advanced semantic indexing.



Semantic Chunking Architecture

~1,450

CLEAN CHUNKS

Token Window Constraints

Slices files into segments of 450–520 tokens. Preserves semantic coherence and paragraph-level context windows.

Hallucination Guardrails

Every semantic chunk is strictly injected with **document-level metadata** (origin, region, authority). This guarantees absolute compliance traceability, preventing high-risk pesticide dosage hallucinations.

Multi-Crop Yield Dataset: Harmonizing 440,962 Historical Records

Preserving Non-Linear Agro-Climatic Dependencies via MissForest Imputation

Dataset Harmonization

4 Disparate Indian Agri Datasets (1997–2024)

Fragmented historical streams integrated into a single unified workspace, resolving inconsistent schema structures, varying temporal metrics, and missing environmental variables.

Mapped to Schema Structure

16-Column Unified Target Schema

Spatial-Temporal	Year, State, District, Coordinates
Crop Metrics	Crop, Season, Cultivated Area, Yield
Agro-Climatic	Rainfall, Temperature, Soil Moisture
Agricultural Inputs	Fertilizer (NPK consumption), Irrigation

Non-Parametric Imputation

Applied MissForest iteratively across the dataset. The non-parametric random forest approach handles mixed-type variables without parametric distribution assumptions.

Preserves complex rainfall, temperature, and fertilizer interactions.

440,962

Imputed Rows

124

Unique Crops

35

States & UTs

Agent Tool Readiness

Exposes the fully completed, schema-mapped agro-climatic dataset directly to the LLM agent via specialized Python data execution APIs.

ReAct LLM



Python Tool



Deterministic Exec

Standardized Preprocessing Across All Modalities

Engineering Uniform, Noise-Free Inputs Across Vision, Text, and Tabular Streams

MODALITY	PREPROCESSING PIPELINE	TECHNICAL OUTCOME
 Vision (Images)	PREPROCESSING STEPS Aspect-ratio preserving letterbox pad to 256×256 RGB	TECHNICAL OUTCOME  Ready for live 224×224 training crops
 Text (KCC & PDFs)	PREPROCESSING STEPS Indic Unicode NFC normalization, Devanagari script harmonization, and PII removal	TECHNICAL OUTCOME  Eliminate zero-width artifacts
 Tabular (Yield)	PREPROCESSING STEPS Ordinal Encoding and unit standardization (pieces to tonnes)	TECHNICAL OUTCOME  Ensure consistent metric scales

Algorithmic Defenses Against Data Leakage & Evaluation Inflation

Why Our Validation Metrics Will Reflect True Real-World Field Performance



Vision Defense

pHash GroupShuffleSplit guarantees zero near-duplicate overlap across Train, Validation, and Test sets.



Yield Defense

Temporal train/test splitting enforced before modeling to eliminate time-series lookahead leakage.



RAG/Metadata Defense

Strict Uttar Pradesh geographical filtering prevents cross-state agronomic advice.

Milestone 2 Final Deliverables & Training Readiness Verification

Confirming 100% Readiness for Immediate Milestone 3 Model Training



Vision Training Manifests

VALIDATED

TRAINING READY PARTITIONS

- train_manifest.csv
- val_manifest.csv | test_manifest.csv
- frozen label_to_idx.json

STATUS: Balanced datasets locked for vision training



KCC Vector RAG JSONL

1.46M CHUNKS

VECTOR DATABANK STRUCTURE

- File: kcc_chunks_rag.jsonl
- Size: 1.18 GB total database volume
- Processing: Fully tagged and verified chunks

STATUS: Indexed and optimized for RAG retrieval



PDF Knowledge Base

170 SOURCES

KNOWLEDGE RETRIEVAL SOURCE

- File: pdf_chunks.jsonl
- Scope: 170 authoritative domain-expert PDFs
- Format: Fixed-width 512-token segments

STATUS: Standardized embedding chunks verified



Yield Prediction Dataset

440,962 ROWS

TABULAR MODELING ENGINE

- File: production_unified_imputed.csv
 - Row Count: 440,962 complete records
 - Quality: Zero-null imputation completed
- STATUS: Fully standard-scaled & regression-ready

Roadmap to Milestone 3: Model Training, Vector Indexing & Agent Integration

Sprint Plan for Building the Live AgriAssist Prototype

SPRINT 1: BASELINE MODELS

Fine-tune EfficientNet-B0 on the Vision 20-class dataset; embed and index KCC/PDF chunks into a vector database.

SPRINT 2: AGENTIC INTEGRATION

Connect Decoupled Vision Router to ReAct LLM Agent; expose predict_yield function calling tool.

SPRINT 3: FIELD ROBUSTNESS

Evaluate end-to-end latency (targeting under 3 seconds) and stress-test Vision against held-out PlantDoc field imagery.

Architectural & Preprocessing Enhancements Under Evaluation

High-Priority Engineering Optimizations Considered for Milestone 3 Polish (Referencing Milestone 2 Recommendations Matrix P2-P5)

VISION API

Structured Output & TTA Abstention

Rec P2

- **Test-Time Augmentations (TTA):** Apply 4–8 transformations at inference to measure predictive entropy across views.
- **Structured Differential JSON:** Emit top-2 disease probabilities to let the ReAct Agent ask clarifying dialogue rather than guessing on blurry images.

KCC VECTOR DB

MinHash LSH Near-Duplicate Index

Rec P3

- **Locality Sensitive Hashing:** Run MinHash (3-grams) across 1.46M KCC chunks within isolated (Crop, QueryType) metadata buckets.
- **Index Compression:** Deduplicate colloquial rephrasings down to ~400k canonical chunks, achieving a **~70% VRAM & latency reduction**.

RAG KNOWLEDGE BASE

Parent-Child Table Indexing

Rec P4

- **Fine-Grained Child Chunks:** Index granular segments (~150 tokens/table rows) in MuRIL vector space for highly precise similarity retrieval.
- **Parent Section Recontextualization:** Dynamically map retrieved nodes to parent context (~600–1,000 tokens) to preserve dosage schedules.

VISION BACKBONE

Hierarchical Multi-Task Head

Rec P5

- **Coarse-to-Fine Classifier:** Replace flat 20-class softmax with a hierarchical head resolving Crop Species before Specific Disease classification.
- **Logical Boundary Enforcement:** Structurally guarantees that a Rice image can never be misclassified with a Wheat disease label.